



The fine print of collaboration: How contractual provisions govern IP and disclosure in publicly funded research

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ABSTRACT

We investigate how public funding and competing orientations shape the contractual provisions regarding intellectual property (IP) and disclosure in university–industry collaborations. By systematically coding and analysing the contracts of 484 collaborations funded by the Research Council of Norway, we identify considerable heterogeneity in IP ownership, exploitation rights, publication policies, and confidentiality rules. Our cluster analysis reveals three distinct governance models: proprietary partnerships, controlled access agreements, and open science collaborations. Controlled access agreements—where universities retain IP ownership, but industry partners hold exploitation rights and impose disclosure restrictions—are the most common. Regression analyses show that higher public funding shares are significantly associated with reduced industry ownership and use rights, but not with publication or confidentiality restrictions. Moreover, projects with more diverse consortia and research-oriented funding instruments tend to adopt less restrictive disclosure provisions. These findings challenge the simplistic open–closed dichotomy and offer a nuanced understanding of how contractual provisions mediate the diffusion of publicly funded research. Our study contributes to debates on open science, knowledge monopolies, and the strategic governance of research outcomes.

1. Introduction

Collaborative research between universities and industry has become increasingly prevalent, serving as a crucial mechanism for knowledge transfer, driving innovation, and ultimately fostering economic growth (Bruneel et al., 2010; Cohen et al., 2002; Perkmann et al., 2013; Szücs, 2018). Often supported by public funding (OECD, 2019), a key question in these partnerships concerns the governance of the projects' outcomes, typically defined in contractual agreements. These agreements regulate the creation and exploitation of intellectual property rights as well as procedural details of the project, such as resource commitments from collaborating organisations and confidentiality provisions (Gans et al., 2017; Gersdorf et al., 2019). Prior literature on the sponsorship of academic research suggests that “public” sponsors, such as state agencies, focus on advancing public knowledge, expecting outcomes to be disclosed through various channels, including publications, whereas “private” sponsors, such as industry, focus on extracting rents from outcomes by restricting public disclosure (Czarnitzki et al.,

2015b; Gans and Murray, 2012). In fact, governments worldwide seek to enhance the impact of public research funding by targeting university–industry collaborations, providing incentives for universities to consider their “third mission”, which is a contribution to addressing societal needs rather than only education and pure (basic) research (Etzkowicz and Leydesdorff, 2000). However, our understanding of how public sponsorship of collaborative research projects shapes the negotiations of contractual provisions among the partners regarding the allocation of intellectual property (IP) rights and disclosure of research outcomes is limited.

The extant literature on university–industry collaboration would lead us to believe that competing orientations—differences in goals and expectations between collaboration partners—are pervasive in these partnerships (He et al., 2021; Tartari et al., 2012). Universities and the academic scientists involved typically seek to create scientific knowledge that can be broadly disseminated. In contrast, firms aim to profit from marketable products and services that require research results to remain secret or protected through IP rights (Sauermann and Stephan,

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2013). Given the competing orientations of the partners involved, it is unclear how public sponsorship plays out in negotiating the collaboration contract.

Studying the contractual provisions in university-industry collaborations provides an understanding of the intentions of the collaborating parties and the safeguards that reduce behavioural uncertainty, facilitate trust between partners, and create an understanding of how the outcomes can be used in a mutually acceptable and beneficial way (Bstieler et al., 2014; Olaisen and Revang, 2017). Although the contracts are inherently “incomplete” due to various informational challenges, they represent negotiated agreements among the partners that can significantly influence the conduct and nature of the research. Whereas prior literature often counterpoints “open science”, emphasising rapid public disclosure of new knowledge, with closed practices characterised by secrecy and IP (Dasgupta and David, 1994; Mukherjee and Stern, 2009), the partners may agree on a wide variety of IP and disclosure approaches. Selective openness, strategic ambiguity, and conditional access that often characterise real-world projects will be overlooked if the analytical framework is based on a simple dichotomy. Hence, in this research, we ask: *How do the partners in university-industry collaborations govern IP and disclosure of the project results?* More specifically, *what are the patterns of contractual provisions regarding the allocation of IP and disclosure, and how does the degree of public funding associate with these patterns?*

We focus our analysis on four types of contractual provisions (Egelie et al., 2019), governing (i) the *ownership* of foreground IP, defined as all knowledge produced during the tenure of a collaborative project (Granstrand and Holgersson, 2014), (ii) the exploitation or *use rights* to the foreground IP, (iii) the ability to *publish* project results, and (iv) the *confidentiality* or secrecy of project results. We study these provisions based on the collaboration agreements of 484 research projects involving university and industry partners. The projects were funded by the Research Council of Norway (RCN) between 2009 and 2017 and are in all fields of science. In the Norwegian public funding system for research, the RCN provides funding for basic and applied research as well as collaborative projects (Borlaug et al., 2019). Norway is not a member of the European Union (EU) but participates in the EU single market and the EU Framework Programmes for Research and Technological Development, making the RCN comparable to other European state research funding agencies.

In a comprehensive process, we code all contracts according to the four types of IP and disclosure provisions. This allows us to draw a fine-grained picture of how the contractual negotiations between the collaboration partners have played out beyond a simple distinction between open and closed practices. In the second step, we collect data on the characteristics of the projects and participants and relate the share of public funding received, as well as project and participant characteristics, to the observed types of contractual provisions. In addition, we collect evidence in the form of interviews. Specifically, we ask seven project managers and participants about the outcomes and governance of selected projects from our dataset. These interviews complement the analysis of the contracts by focusing on the actual outcomes of the projects, how they relate to the initially agreed contract, and how the negotiation and possible re-negotiation dynamics unfolded.

Our analysis uncovers three distinct governance models that reflect nuanced contractual arrangements adopted by the consortia, challenging a simple open-closed dichotomy. While a substantial share of the projects in our sample can be characterised as “proprietary partnerships” or “open science collaborations”, most projects represent “controlled access agreements”, with IP ownership allocated to universities and exploitation rights allocated to industry, combined with restrictions regarding publication and confidentiality of the project results. Notably, higher public funding shares correlate with a lower degree of industry ownership and exploitation rights but not with publication and confidentiality restrictions. Proprietary partnerships (controlled access agreements) exhibit a share of public funding below

(above) the average.

We contribute to the extant literature on university-industry collaborations in at least two ways. First, our research takes its point of departure in a project's collaboration agreement, which represents the negotiated intentions of the project participants regarding how the research results shall be governed. Building on prior research that characterises collaboration agreements of publicly funded research projects as complex and entwined legal provisions governing both the IP on and disclosure of research results (Bstieler et al., 2014; Egelie et al., 2019), we provide a novel perspective on the diversity of contractual provisions as a key mechanism determining the extent to which publicly funded research results diffuse into the public domain (or not). Our findings contribute to the ongoing debate about open science when universities engage with industry (e.g., Czarnitzki et al., 2015b; Mukherjee and Stern, 2009) and demonstrate a new method for evaluating collaborative research projects ex-ante. In fact, despite the growing concern about knowledge monopolies arising as a result of concentrated ownership and secrecy (Rai and Cook-Deegan, 2017), identifying their emergence in publicly funded research projects is challenging. Collaboration agreements, hence, provide an early view on the unfolding IP and disclosure practices.

Second, we contribute to the literature on the public sponsorship of research (Gans and Murray, 2012; Szűcs, 2018), focusing on how public funding associates with the collaborative projects' negotiated IP and disclosure provisions. We go beyond simple assertions of public sponsors promoting and private sponsors restricting disclosure and show how the public funding share interacts with competing orientations of the partners involved (He et al., 2021; Sauermann and Stephan, 2013) in shaping a nuanced collaboration agreement. In that sense, although our results are based on a specific country context and funding agency only, they can inform policymakers on how to align public funding with policy objectives in the development of policy instruments, the resulting contractual models, and the follow-up of project portfolios (Howoldt and Borrás, 2023).

The remainder of the manuscript is organised as follows. Section 2 presents background literature and our analytical framework, discussing the four types of contractual provisions governing a project's IP and disclosure practices. Section 3 presents our research design and methods, while the results are presented in Section 4 and discussed in Section 5. We conclude by outlining avenues for future research.

2. Literature background

2.1. Sponsorship of research and its implications for contractual agreements

The sponsorship of research, whether academic research conducted at universities or collaborative research conducted as a joint project between universities¹ and industry, usually entails more than just the transfer of funds. It involves contractual agreements that specify ownership, exploitation rights, as well as the disclosure and dissemination of research findings, methods, or materials (Gans and Murray, 2012; Geuna, 2001). David (2004) highlights how sponsorship historically contributed to transforming scientific norms and structures from secrecy-driven systems to the modern open science framework characterised by the rapid public disclosure of knowledge. Open science is widely regarded as an efficient system for producing cumulative,

¹ In the following, we will use the term universities as shorthand for all types of publicly funded research institutions including universities, university hospitals, and research institutes. In the Norwegian context, research institutes may be foundations, limited companies owned by public bodies, or government entities, such as the Norwegian Veterinary Institute. They are publicly funded applied research organisations and conceptually close, but legally distinct from universities (Gulbrandsen, 2011).

accessible, and reliable public scientific knowledge, fostering societal welfare (Dasgupta and David, 1994; Mukherjee and Stern, 2009).

In this sense, these contracts often reflect the goals and institutional reward systems of sponsors, which can significantly shape norms and incentives, and thus impact the conduct and nature of the research. Public sponsors, such as science-focused state agencies or private foundations, typically prioritise advancing public knowledge through disclosure, aligning with open science norms. The expectation of publicly disseminated findings, often demonstrated through publications, reinforces these norms and secures continued support. In contrast, private sponsors, such as industry, may focus on deriving economic benefits by imposing restrictions on knowledge disclosure, which in turn undermines open science principles (Czarnitzki et al., 2015b).

Based on a theoretical framework to explore how public and private sponsorship of academic research affects openness and project selection, Gans and Murray (2012) analyse research agreements at 20 major U.S. universities and show considerable variation in contractual provisions regarding publication restrictions. Among 16 universities with such clauses, prepublication review periods and delays differed significantly. Most agreements permitted industrial sponsors to review and restrict confidential information disclosure, though five university contracts allowed full disclosure. In a similar vein, Blumenthal et al. (1996b) surveyed 210 life science companies, finding that 58 % required researchers to maintain confidentiality for over six months. Thursby and Thursby (2007) report that 90 % of contracts between universities and private firms included publication delay clauses. These findings are consistent with studies asking academic researchers directly. Academics working on an industry-sponsored project are more likely to face publication restrictions, including delays, bans, and prepublication reviews (e.g., Blumenthal et al., 1986; Blumenthal et al., 1996a; Czarnitzki et al., 2015b), to deny others access to research inputs such as materials and data (Campbell et al., 2000; Campbell et al., 2002; Czarnitzki et al., 2015a), and to become more secretive to protect proprietary interests (Hong and Walsh, 2009). In sum, these studies suggest that scientists with private sponsorship are more likely to encounter disclosure restrictions than those without because the terms of the sponsorship agreements bind them.

Besides academic research conducted at universities, public and private sponsorship are also important funding sources for collaborative research involving universities and industry partners. Governments worldwide have increasingly been targeting university-industry collaboration to provide incentives to engage with one another, transfer knowledge, and jointly work on problems requiring the partners' skills and expertise (Nugent et al., 2021). Indeed, university-industry collaboration has frequently been shown to benefit both academic research at universities (e.g., D'Este and Perkmann, 2011; Perkmann and Walsh, 2009; Perkmann et al., 2021) and firm innovation (e.g., Mansfield, 1995; Murray, 2002). Because of these benefits, public sponsors typically also accept collaboration agreements that allow the partners to restrict the disclosure of research results and to exploit the IP generated commercially (Egelie et al., 2019). Therefore, public sponsorship of collaborative research may misalign with the norms of open science. Further, these contracts are incomplete, in that they concern research with an unknown outcome. Thus, they may need to be renegotiated as the research develops toward utilisation (Holgersson et al., 2022).

2.2. Competing orientations in collaborative projects

Beyond the sponsor's expectations, university-industry collaborations are characterised by competing partners' orientations. Despite shared aspirations for the success of such collaborations, universities and firms may pursue distinct organisational goals and have distinct cultures, referred to as institutional logics—academic and commercial—informing their values, practices, and motivations for collaboration (Perkmann et al., 2013; Sauermann and Stephan, 2013). While firms aim to profit from marketable products and services, universities

typically focus on generating novel scientific knowledge (Merton, 1973). Industrial research tends to emphasise short-term performance, contrasting with the long-term exploratory orientation of academic research (Meyer-Krahmer and Schmoch, 1998; Tartari et al., 2012). These differences are particularly evident in reward systems (peer recognition vs. financial returns) and attitudes toward knowledge disclosure (public dissemination vs. intellectual property protection; Gans et al., 2017).

Academic researchers, compared to their corporate counterparts, often approach collaborations based on personal preferences and values, viewing them as opportunities to benefit subsequent research (Roach and Sauermann, 2010; Sauermann and Stephan, 2013). While corporate scientists also value openness and may even accept reduced pay in exchange for publication opportunities (Gans and Murray, 2012; Stern, 2004), differing norms and incentives lead to varying priorities regarding the disclosure and commercialisation of collaborative outcomes. In that sense, academic scientists typically prioritise journal publications more than their corporate collaborators (Estrada et al., 2016).

In recent decades, however, the historically clear distinction between academic and commercial institutional logics has become increasingly blurred. Secrecy and limited support for open science are no longer exclusive to companies (Perkmann et al., 2013). Since the implementation of the Bayh-Dole Act in the U.S. and similar policies in Europe, universities increasingly prioritise research commercialisation, which requires control over the IP, often through patents or confidentiality.² A notable example is the controversy over essential patents related to the CRISPR-Cas9 technology between the University of California, Berkeley, and the Broad Institute of MIT and Harvard—universities that have developed commercial interests, including equity stakes in start-ups commercialising CRISPR applications under exclusive licenses (Egelie et al., 2016; Rai and Cook-Deegan, 2017). It is striking that CRISPR research has largely been funded by the U.S. National Institutes of Health, socialising the risks and costs of this research through public funding while privatising the resulting financial gains (Egelie et al., 2018).

The CRISPR case illustrates the tension between universities' ambition for the dissemination of knowledge production and the more controlled flow of knowledge when such a platform technology is commercialised. This tension will likely influence the negotiation dynamics of the collaboration contract. Relatedly, the current debate at the EU level on Open Strategic Autonomy highlights how funding agencies and policymakers need to balance knowledge diffusion and openness with concerns about technological sovereignty and control over research outputs (Edler et al., 2023), requiring governance at the level of research collaborations and research project portfolios.

In the following, we first seek to identify and describe the various contractual arrangements that can be found in university-industry collaboration and, second, to relate these arrangements to the characteristics of the partners and the project itself, with a particular focus on the extent to which a project receives public funding.

3. Methods

3.1. Data

Under a confidentiality agreement, we were granted access by the Norwegian government to the collaboration contracts of university-industry collaborative projects funded by the RCN, the foremost Norwegian public research sponsor. The RCN provides funding for basic and

² The legal threshold for establishing trade secrets are in brief i) secrecy, that is no publishing, ii) a commercial value from being secret and iii) measures to keep it secret. This threshold is challenging to demonstrate in a university context.

applied research as well as collaborative projects (Borlaug et al., 2019), similar to agencies like UK Research and Innovation (UKRI). The RCN allocates funding through competitive calls for proposals, where applications are evaluated by international experts, as is common practice among research funding agencies worldwide.³ While Norway is not a member of the European Union (EU), it participates in the EU single market and the EU Framework Programmes for Research and Technological Development, aligning with the practices of other national funding bodies that promote international research partnerships. Applicants from Norway have been highly successful in securing grants from Horizon Europe (2021–2027), which reflects the high quality of their research proposals and participants.⁴ Similar to other Nordic countries, Norway has a high level of public funding for research and development (R&D). However, the business sector's expenditure for R&D is lower, at about the EU's average.⁵ Overall, we conclude that Norway and the RCN provide a setting that is comparable to other countries, suggesting that the results reported here can inform policy-making in different contexts.

Out of 19,644 projects funded by the RCN between 2009 and 2017 with a total budget of NOK 64.1 billion (about EUR 5.5 billion), we randomly selected an initial sample of 1000 collaborative projects that involved at least one university and one company. We dismissed projects without archived, readable contracts and ended up with a final sample of 484 projects.

Funding allocations from the RCN are always carried out according to a contract that sets out the RCN's general terms and conditions for the use of the research funding and clarifies the parties' rights and obligations regarding the implementation of the project. Hence, before the project owner, usually the coordinator, signs the contract, the parties negotiate and agree on the contractual provisions that govern IP and disclosure. Besides the contract, we gather information on the project and the participants involved, provided to us by the RCN. Comparing the project budget and the number of participants involved in these projects with those we discarded, we find no significant differences.

In addition, we conduct seven interviews with project managers and participants from four projects in our dataset after they had concluded. We select these projects based on the diversity of contractual provisions found. The semi-structured interviews were performed in 2024 by two of the authors. In the interviews, we sought to understand the outcomes of the projects and how the partners eventually decided to govern them, whether and to what extent the collaboration agreement was changed, and how they handled the practical matters regarding IP and disclosure during the project period. The interview guide can be found in Table A2 in the appendix. Each interview lasted about 1 h. We complement the interview data by investigating patent and publication outputs using databases maintained by the RCN and technology transfer offices at universities.

3.2. Coding IP and disclosure provisions

To identify the contractual provisions regarding IP and disclosure, we conducted an ontological analysis of contractual models (Bullinger, 2009; Rosemann et al., 2004). These models included the template used by the RCN, the DESCA contract template for EU-funded research consortia and the UK equivalent “Lambert toolkit” (DESCA, 2017; Eggington et al., 2013). The collaboration agreements were reviewed by two authors, focusing on the four types of contractual provisions chosen:

ownership of the foreground IP, exploitation or use rights to the foreground IP, publication restrictions, and confidentiality provisions. The templates for collaboration agreements, established by most funding bodies, are typically used by the partners when the contract is negotiated, even though they are not mandatory. About 60 % of the projects in our sample used the RCN template, and the remainder made few modifications to the template. Importantly, the consortium contracts are agreed upon ex-ante, i.e. before the collaborative project has started. While the contracts allow us to study the partners' intentions regarding the governance of project outcomes, they do not provide information on the actual outcomes. We follow up on the actual project outcomes through the interviews with selected projects described further below.

Provisions regarding the ownership of the foreground IP include the definition of ownership of the results from within the project's scope. Ownership refers to the ability to control and manage access to the IP. Provisions regarding exploitation or use rights to the foreground IP define the rights to use the results within the project's scope. Use rights define the opportunities for commercial utilisation of the project results, both exclusively and non-exclusively, and the right to use the IP for further research. Provisions regarding the publication of research results relate to the various degrees of banning or delaying publication, for example, to enable the filing of patents. Finally, confidentiality provisions relate to how research results should be kept confidential and for how long. In that sense, provisions regarding ownership and use rights govern the control over the IP on the research results, whereas publication and confidentiality provisions reflect different degrees of the projects' openness. Obviously, these types of provisions are not independent of each other. By default, ownership implies the right to use a project result, but the contract may also specify a different allocation of ownership and use rights to the involved parties. Moreover, provisions regarding publication and confidentiality are directly linked, but can reflect various aspects of the information that is allowed to be publicly released.⁶ We code these as separate types because they appear separately in the collaboration contracts. In our conversations with the RCN, we learned that universities seek to explicitly clarify publication opportunities while companies seek to specify confidentiality rules.

In addition to the four types of provisions that we use in our study, we coded four additional types of provisions based on the ontological analysis of contract models. These concern background IP, defined as IP that exists and is brought into a collaborative project by one of the partners involved (Granstrand and Holgersson, 2014), use rights for research and education, liabilities, and warranties. While these constitute essential parts of the contract, clarifying IP governance and the obligations of the partners, we found virtually no variation in how these provisions were handled in the projects. In that sense, all contracts state that the background IP belongs to the contributor, that universities and research institutes are granted the right to use the results in further research and education, and that the parties shall not be responsible for any indirect or consequential losses because of a breach of the contract. Thus, we decided not to consider these provisions further. In sum, the contracts constitute a bundle of rights, i.e. a collection of individual rights defined in the contractual provisions that can be separated and transferred individually among the parties or held collectively.

In the first step, 30 contracts were independently coded by two authors, who then compared their coding, reviewed differences, and agreed on a common protocol. For each of the four types, five distinct values were identified. Since most of the projects use the contract model by the RCN that provides example formulations for certain provisions, the codes involved little disagreement. The remainder of the contracts was then divided among the two authors and coded independently.

Table 1 shows the results of coding the IP and disclosure provisions

³ See <https://www.forskningsradet.no/en/financing/what/> (accessed: 9 December 2024).

⁴ See <https://www.forskningsradet.no/en/news/2024/norway-on-the-top-europe-in-research-funding-from-the-eu/> (accessed: 9 December 2024).

⁵ See [https://ec.europa.eu/eurostat/statistics-explained/index.php?title=R%26D_expenditure#:~:text=While%20R%26D%20expenditure%20in%20the,in%20Switzerland%20\(2021%20data\)](https://ec.europa.eu/eurostat/statistics-explained/index.php?title=R%26D_expenditure#:~:text=While%20R%26D%20expenditure%20in%20the,in%20Switzerland%20(2021%20data)) (accessed: 9 December 2024).

⁶ Practical examples are proprietary code becoming Open Source, confidential images or data documenting research being published under Creative Commons as well as patent applications under work.

Table 1
Coding of the IP and disclosure provisions.

	Ownership	Use rights	Publication	Confidentiality
5	Industry partners own all IP on project results. ($n = 69$)	Industry partners have exclusive use rights to all IP on project results. ($n = 32$)	Project results may not be published. ($n = 7$)	All information is confidential if not already public; no time or other limitations. ($n = 13$)
4	A specific partner owns the IP on project results based on own background, dominating contribution or commercial interests; otherwise joint ownership. ($n = 66$)	A specific partner has exclusive use rights to IP within specified fields of use; otherwise the partners have non-exclusive, royalty-free use rights to IP in all fields. ($n = 128$)	Project results may be published; publications can be delayed (e.g. due to patenting concerns). ($n = 106$)	All information is confidential if not already public; limited in time. ($n = 150$)
3	Joint ownership of all IP on project results. ($n = 55$)	All partners have non-exclusive, royalty-free use rights to all IP on project results. ($n = 216$)	Project results shall be published; publications need approval from steering committee which could object and request modifications before publication. ($n = 220$)	All information is confidential if marked and justified; limited in scope and time. ($n = 250$)
2	When partners have worked jointly on project results and their respective contribution cannot be determined, partners have joint ownership; otherwise a specific partner owns the IP on project results. ($n = 283$)	All partners have non-exclusive, royalty-free use rights to relevant IP during the project period; further use rights can be negotiated upon request. ($n = 104$)	Project results must be published; publications can be delayed (e.g. due to patenting concerns). ($n = 138$)	Project partners must specifically request confidentiality; limited in time and conditional on approval by steering committee. ($n = 54$)
1	University partners own all IP on project results. ($n = 11$)	University partners have exclusive use rights to all IP on project results. ($n = 4$)	No publication restrictions; project results must be published. ($n = 13$)	No confidentiality provisions. ($n = 17$)

found in the collaboration contracts of the 484 projects, together with the frequency of how many times a certain provision has been observed in the contracts. Regarding the ownership of the foreground IP, we find that the most prevalent provision ($n = 283$) is joint ownership for joint work unless the project results can be clearly traced back to a specific partner, in which case this partner holds sole ownership. We also find that contracts often stipulate that industry partners own all the IP ($n = 69$) and that a specific partner owns the IP on project results based on their own background, dominating contribution or commercial interests ($n = 66$), which can be interpreted as industry-oriented. In contrast, joint ownership ($n = 55$) and university ownership alone ($n = 11$) are much less commonly used.

Concerning the exploitation or use rights to the foreground IP, we find non-exclusive, royalty-free use rights to all IP by all project partners

to be the most prevalent provision ($n = 216$). However, we also find a sizable number of projects adopting industry ($n = 32$) or industry-oriented ($n = 128$) provisions regarding use rights. Under these provisions, industry partners either have exclusive use rights to all IP or to IP within specific fields, indicating commercial opportunities in certain markets. Another frequent provision is non-exclusive, royalty-free use rights during the project period only, after which renegotiations need to take place ($n = 104$). As expected, we only find four cases where university partners have exclusive use rights. We also note that the contracts do not include a more precise description of different types of licenses, since the delineation of the project outcomes is unknown *ex ante*.

Regarding publication and confidentiality restrictions, we find an intermediate level of restrictions to be most common, with 220 and 250 cases, respectively. Here, project partners can request publication but require pre-approval, and information can be marked as confidential but only for a limited time. For both types of provisions, we also find sizable numbers of contracts with slight variations in the level of restrictions on openness. In contrast, a full publication ban ($n = 7$) or no publication restrictions at all ($n = 13$), as well as full confidentiality ($n = 13$) or no confidentiality at all (17) are rather infrequent provisions.

In sum, the coding of contractual provisions shows a remarkable diversity of contractual provisions that goes far beyond simple characterisations of closed versus open science. While the five categories of each variable are distinct ways in which the provisions in the contracts are shaped, they indicate an increasing degree of industry orientation, which our conversations with project managers, technology transfer offices and contract lawyers confirm. Table 1 shows the ordering, assigning a value of 1 to provisions that favour ownership and use rights assigned to universities and that do not impose any publication or confidentiality restrictions. The value of 5 is given to provisions that assign ownership and use rights to industry and that impose broad publication and confidentiality restrictions for project results.

Consequently, we use these four variables, i.e. the degree of industry ownership, the degree of industry use rights, the degree of publication restrictions, and the degree of confidentiality restrictions, as our main dependent variables. In addition, we calculate two index variables, summing up the values of the degree of industry ownership and industry use rights, as well as the degree of publication and confidentiality restrictions, in order to account for the interrelated nature of the concepts. As a robustness check, we construct two dummy variables that take the value of 1 for the highest degree of industry ownership and industry use rights, respectively, and 0 otherwise. Because of the low number of observations for the highest degree of publication and confidentiality restrictions, we use both the highest and second highest degree of restrictions to construct dummy variables.

3.3. Explanatory variables

3.3.1. Public funding share

We define the public funding share as the amount contributed to the project by the RCN as a share of the total project budget.

3.3.2. Control variables

We include the total number of partners involved in the project and the concentration of the different types of partners to control for the composition of the consortium. Specifically, we calculate a Herfindahl-Hirschman measure of concentration based on the shares of companies, universities, research institutes, as well as public bodies and other partners in the consortium. A value of 1 of the concentration measure would indicate that the consortium only consists of one type of partner (which is not the case in our sample since we investigate collaborative projects), while a lower value indicates less concentration. Moreover, we include the total budget (millions of NOK in log.) to control for the size of the project, and we include a dummy variable for the funding instrument type (research, with commercial and other

serving as reference categories), as defined by the RCN. The RCN funds projects in all fields of science. Hence, we control for fields by including dummy variables indicating whether a project was in biotechnology, energy, engineering/ICT, ocean, or another field, with biotechnology being the reference category in the regression analyses. Finally, we include year dummies for the years the projects started.

3.4. Data analysis

Our empirical analysis unfolds in several steps, following our research questions. After showing descriptive statistics on project characteristics, we estimate ordinary least squares (OLS) regressions for each of the four dependent variables measuring the contractual provisions that we treat as approximately continuous. To account for the interrelationship between the dependent variables, we use the index variables for the degree of industry ownership and use rights and the degree of publication and confidentiality restrictions in a seemingly unrelated regression (SUR) model as well as in an OLS regression, in which the degree of publication and confidentiality restrictions serves as the dependent variable and the degree of industry ownership and use rights as the explanatory variable. To account for heteroscedasticity, we use standard errors clustered by 45 RCN funding programs to which the projects in our sample belong. Moreover, we use robust, heteroscedasticity-consistent standard errors as a sensitivity check which yields quantitatively very similar results (available from the authors upon request). As a robustness check, we estimate probit models using the dummy variables for a high degree of industry ownership, industry use rights, publication restrictions, and confidentiality restrictions.

In the next step, we seek to uncover patterns of contractual provisions that commonly occur together among the projects in our sample. For this purpose, we run a hierarchical cluster analysis using Ward's linkage, a method that focuses on minimizing the increase in the "error sum of squares" when merging clusters, and the Euclidean distance measure. Ward's linkage aims to find groups of observations that are as internally similar (with respect to variance) and as distinct from other groups as possible (Everitt et al., 2011). Subsequently, we describe the cluster solution and run probit regressions explaining the cluster solution using the explanatory variables as before.

Finally, we report the results from the seven semi-structured interviews using a simple thematic approach (e.g., Braun and Clarke, 2006). The interviews were transcribed and reviewed to identify recurring topics and perspectives relevant to the main research questions. Coding was done manually and iteratively, focusing on capturing patterns in how participants described their experiences. The analysis aimed to identify key themes that could help contextualise and enrich the interpretation of the quantitative findings.

4. Results

4.1. Descriptive statistics

Table 2 shows descriptive statistics on the projects and the participants. In total, the 484 collaborative projects involve 3944 partners. We find that the RCN on average provides funding for 59 % of the projects' total budget. The RCN fully funds 23 projects. Moreover, we find that, on average, eight partners are involved in the projects, with a maximum of 40 partners. The value of the concentration measure indicates a moderate degree of concentration, i.e. projects are typically dominated by one partner type. In fact, about half of the partners in each project are, on average, companies. 17 % of all projects are funded through an instrument that can be characterised as research-oriented. Most of the projects (33 %) fall into the field of biotechnology, followed by energy (29 %), ocean (13 %) and engineering/ICT (11 %). Finally, most of the projects in our sample (50 %) are from the most recent years 2015–2017.

Fig. 1 shows the development of the degree of industry orientation

Table 2

Descriptive statistics ($n = 484$).

Variable	Mean	Std. dev.	Min	Max
Degree of industry ownership	2.791	1.158	1	5
Degree of industry use rights	3.165	0.867	1	5
Degree of publication restrictions	2.909	0.814	1	5
Degree of confidentiality restrictions	3.182	0.799	1	5
Degree of industry ownership and use rights	5.957	1.721	2	10
Degree of publication and confidentiality restrictions	6.091	1.330	2	10
RCN funding share	0.593	0.220	0.085	1
Number of partners	8.149	5.107	2	40
Partner concentration	0.484	0.124	0.256	0.900
Project type: research (d)	0.174	0.379	0	1
Total budget (lnNOK)	9.635	1.120	5.298	12.973
Biotechnology (d)	0.335	0.472	0	1
Energy (d)	0.289	0.454	0	1
Engineering/ICT (d)	0.112	0.315	0	1
Ocean (d)	0.128	0.335	0	1
Other (d)	0.136	0.344	0	1
Years 2009–2011 (d)	0.099	0.299	0	1
Years 2012–2014 (d)	0.401	0.491	0	1
Years 2015–2017 (d)	0.500	0.501	0	1

(d): dummy variable.

regarding the four types of provisions over time. Disregarding the first year in which projects in our sample started because of a rather low number of observations (2009), we find that the degree of industry orientation of the projects has continuously declined. This association suggests that the RCN has put more emphasis on less restrictive provisions over time. We discuss later if this can relate to a shift in policy or other causes.

Table 3 shows pairwise correlations. We do not identify alarmingly high correlations among the explanatory variables, as evidenced by the low average variance inflation factor (VIF) of 1.29, well below commonly accepted thresholds for multicollinearity concerns (Belsley et al., 1980).⁷

4.2. Regression results

Table 4 presents the results from a series of OLS regressions to investigate the relationship between the RCN funding share and the contractual provisions governing IP and disclosure of the project results. In Model 1, we regress the degree of industry ownership of the foreground IP on the RCN funding share and the set of control variables. As expected, we find a significant negative relationship between the RCN funding share and contractual provisions favouring industry ownership over the foreground IP. In that sense, we do find suggestive evidence of public sponsorship working against industry-oriented ownership as the degree of public funding increases. Moreover, we find that larger projects, in terms of the number of participants, are less likely to imply industry-oriented ownership. Apparently, in these projects, the multitude of participants seek to avoid knowledge monopolies and instead favour more dispersed ownership. The concentration of partner types turns out to be unrelated to the degree of industry ownership. We find few other significant relationships. Projects funded in 2009 to 2011 are more likely to adopt industry-oriented ownership provisions, relative to projects funded between 2015 and 2017, the reference category in the estimations. The latter finding may indicate that the RCN emphasised less industry-oriented ownership in more recent years.

Model 2 shows results for the degree of industry use rights to the

⁷ In a sensitivity check available from the authors upon request, we drop the variable "Total project budget (lnNOK)" which is moderately correlated with the number of partners in a project. The results are qualitatively and quantitatively very similar to the main model results.

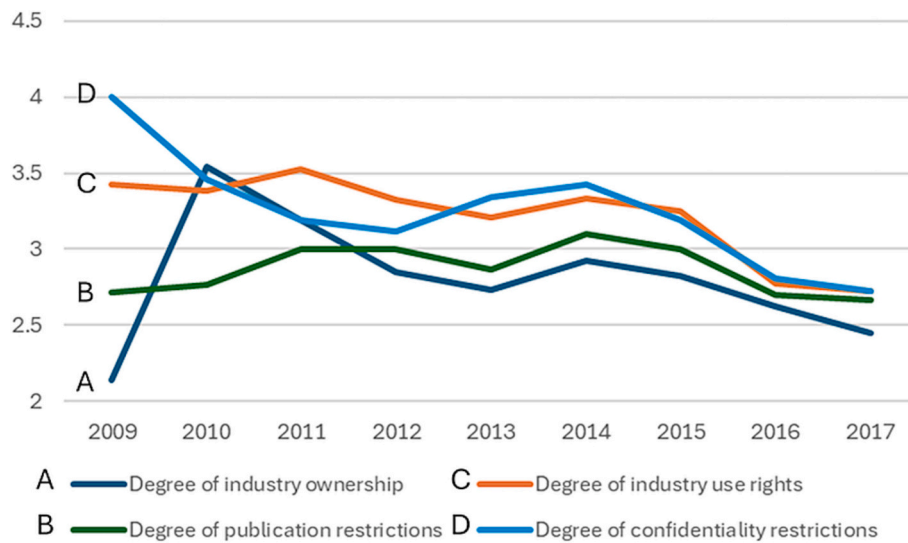


Fig. 1. Contractual provisions over time.

Table 3
Pairwise correlations (n = 484).

		1	2	3	4	5	6	7	8	9	10
1	RCN funding share	1.000									
2	Number of partners	−0.016	1.000								
3	Partner concentration	−0.204	0.139	1.000							
4	Project type: research (d)	0.429	0.062	−0.129	1.000						
5	Total budget (lnNOK)	−0.186	0.470	0.004	0.101	1.000					
6	Energy (d)	0.006	0.148	0.088	−0.172	0.085	1.000				
7	Engineering/ICT (d)	−0.136	−0.106	−0.025	−0.024	−0.018	−0.226	1.000			
8	Ocean (d)	−0.083	−0.117	−0.039	−0.078	−0.085	−0.245	−0.136	1.000		
9	Other (d)	−0.051	−0.019	−0.074	0.088	0.036	−0.254	−0.141	−0.152	1.000	
10	Years 2009–2011 (d)	0.014	0.114	−0.053	0.140	0.246	0.032	−0.008	−0.044	−0.031	1.000
11	Years 2012–2014 (d)	−0.054	−0.113	−0.041	−0.063	−0.086	0.036	−0.116	0.052	0.007	−0.271

(d): dummy variable.

foreground IP. We find largely the same patterns as for the ownership of IP: a negative relationship of the RCN funding share and the number of partners. Moreover, and as expected, we find that projects funded through a research-oriented instrument associate negatively with industry-oriented use rights, while larger projects in terms of total budget show a positive association with industry-oriented use rights. Projects in the energy field are positively associated with industry-oriented use rights, as are projects that were started in earlier years. In sum, Models 1 and 2 show largely consistent relationships between the explanatory variables and the contractual provisions adopted regarding ownership and use rights.

Next, we turn to Models 3 and 4 to investigate the correlates of publication and confidentiality provisions in the contracts. Both models show largely consistent results. Interestingly, we find the RCN funding share to be unrelated to both. Instead, a higher number of partners involved, and projects funded through a research-oriented instrument associate negatively with publication and confidentiality restrictions. For confidentiality restrictions (Model 4), we also find a strong positive association for the concentration of partner types, indicating that a dominant group of partners can advocate for more restrictive confidentiality of the project results. Moreover, increasing project size in terms of total budget turns out to be positively associated with confidentiality restrictions (Model 4). Projects in engineering/ICT (ocean) are positively (negatively) associated with the degree of publication (confidentiality) restrictions in Model 3 (Model 4). Finally, consistent with Models 1 and 2, earlier projects adopt greater publication and confidentiality restrictions.

Table A1 in the appendix shows results from probit regressions in

which the dependent variables take the value of 1 if the degree of industry ownership, use rights, publication, and confidentiality restrictions is high. The results are qualitatively fully consistent with the OLS results. Importantly, we also find that the degree of RCN funding is negatively related to industry ownership and use rights only, but not to publication and confidentiality restrictions.

Table 5 shows the results of the OLS regression in which the variable measuring the degree of industry ownership and use rights serves as an explanatory variable for the degree of publication and confidentiality restrictions (Model 5), as well as the results of a seemingly unrelated regression (SUR) model that estimates both equations predicting the degree of industry ownership/use rights and the degree of publication/confidentiality restrictions simultaneously (Model 6). Model 5 shows a significantly positive relationship between the degree of industry ownership/use rights and the degree of publication/confidentiality restrictions, controlling for all other explanatory variables as before. Moreover, Model 6 shows a positive and significant correlation coefficient of the error terms (ρ) in the two regression equations. Both findings indicate strong interrelations between the two variables, i.e. projects that adopt more industry-friendly provisions regarding ownership and use of the foreground IP also have more restrictions regarding publication opportunities and overall confidentiality of project results. At the same time, the relatively low value of the correlation coefficient ρ indicates that there is considerable heterogeneity among the projects with respect to these provisions, an aspect that the subsequent cluster analysis will explore further.

Table 4

Regression results for the contractual provisions.

	Model 1 – OLS	Model 2 – OLS	Model 3 – OLS	Model 4 – OLS
	Degree of industry ownership	Degree of industry use rights	Degree of publication restrictions	Degree of confidentiality restrictions
Share of RCN funding	–1.788*** (0.282)	–0.689*** (0.202)	–0.056 (0.158)	–0.061 (0.189)
Number of partners	–0.045*** (0.014)	–0.025** (0.011)	–0.018** (0.009)	–0.012* (0.007)
Partner concentration	0.502 (0.404)	–0.361 (0.341)	0.349 (0.283)	0.577*** (0.195)
Project type: research (d)	–0.019 (0.134)	–0.278** (0.118)	–0.392*** (0.090)	–0.225* (0.119)
Total project budget (lnNOK)	–0.061 (0.056)	0.079* (0.044)	0.042 (0.050)	0.140*** (0.030)
Energy (d)	0.097 (0.135)	0.243* (0.134)	0.081 (0.069)	–0.084 (0.064)
Engineering/ICT (d)	–0.229 (0.184)	–0.214 (0.154)	0.179* (0.071)	–0.027 (0.122)
Ocean (d)	–0.147 (0.158)	–0.146 (0.187)	0.065 (0.096)	–0.212** (0.082)
Other (d)	–0.061 (0.206)	–0.083 (0.147)	–0.184* (0.103)	–0.144 (0.091)
Years 2009–2011 (d)	0.383* (0.210)	0.419*** (0.140)	0.07 (0.117)	0.446*** (0.104)
Years 2012–2014 (d)	0.036 (0.099)	0.209* (0.110)	0.143* (0.078)	0.355*** (0.073)
Constant	4.536*** (0.507)	3.101*** (0.554)	2.485*** (0.513)	1.608*** (0.315)
Number of observations	484	484	484	484
R2	0.16	0.14	0.08	0.12
F(11, 44)	17.33	14.01	5.25	6.62
P-value	0.000	0.000	0.000	0.000

(d): dummy variable.

* p < 0.10.

** p < 0.05.

*** p < 0.01.

4.3. Cluster analysis

Based on the Ward's linkage method, we derive the solution from a hierarchical cluster analysis that uses the four variables measuring the degree of industry ownership, use rights, publication, and confidentiality restrictions as cluster variables. To determine the number of clusters, we inspect the dendrogram (Fig. A1 in the appendix) that shows the hierarchical relationships and similarities between a group of observations, with the height of the branches indicating the level of similarity (Everitt et al., 2011). While selecting too few clusters may mask important differences within the clusters, selecting too many clusters complicates interpretation. Hence, we opt for a three-cluster solution that avoids clusters that are too small in size and offers good interpretability.

Table 6 shows descriptive statistics for each cluster that comprise 189, 205, and 90 projects, respectively. We find marked differences between the clusters as the F-test from an analysis of variance (ANOVA) suggests. Projects in Cluster 1 exhibit a rather high degree of industry ownership to the foreground IP as well as above-average values for the other three cluster variables. Projects in Cluster 2 show similarly high value for these three variables but a much lower degree of industry ownership. In contrast, projects in Cluster 3 exhibit low values for all four cluster variables. As a result, we refer to projects in Cluster 1 as “Proprietary partnerships” since these projects have high industry orientation, and we call projects in Cluster 3 “Open science collaborations”. While these two clusters essentially replicate the closed-open dichotomy of collaborative projects between academia and industry, Cluster 2—comprising the largest number of projects—stands out. Projects in this cluster assign ownership to the foreground IP to the university partners but at the same time involve a high degree of control with respect to how and by whom project results can be used, favouring the industry partners in the consortium. Hence, we refer to projects in

this cluster as “Controlled access agreements”.

Table 6 also shows interesting and significant differences among the three clusters for the other explanatory variables. Both, open science collaborations and controlled access agreements, feature the highest degree of public funding (64 %), while it is considerably lower for proprietary partnerships (53 %). Whereas the high degree of public funding is not surprising for open science collaborations, the remarkably high degree of public funding for controlled access agreements suggests that the RCN is willing to fund projects that benefit mostly the industry partners involved as long as the university partners hold ownership to the IP, i.e. the resulting knowledge is not broadly disseminated but kept among the project partners. We also find that open science collaborations exhibit the highest number of partners involved (9.04), immediately followed by controlled access agreements (8.69). Proprietary partnerships have the lowest number of partners involved (7.14). These projects have the highest concentration of partners, i.e. they are typically dominated by companies. Finally, we can see that open science collaborations become more frequent in the later years of our sample, while projects in the earlier years are typically proprietary partnerships.

Table 7 shows the results of probit regressions for the membership in one of the three clusters. While the probit results largely mirror the descriptive analysis, they reveal a number of additional insights. In that sense, we find that the degree of public funding has a significantly negative association with the likelihood that projects are proprietary partnerships, a significantly positive association with controlled access agreements, but no relationship with open science collaborations. This finding shows that the RCN accepts a rather high degree of access restrictions as long as the ownership to the IP lies with universities. The insignificant relationship between public funding and open science collaborations mirrors the findings of Table 4, which did not substantiate a significant association between public funding and publication and confidentiality restrictions. Moreover, projects featuring controlled

Table 5
Regression results for the contractual provisions.

	Model 5 – OLS	Model 6 – SUR	
	Degree of publication and confidentiality restrictions	Degree of industry ownership and use rights	Degree of publication and confidentiality restrictions
Degree of industry ownership and use rights	0.184*** (0.027)		
Share of RCN funding	0.340 (0.272)	−0.689*** (0.202)	−0.056 (0.158)
Number of partners	−0.016 (0.012)	−0.025** (0.011)	−0.018** (0.009)
Partner concentration	0.900** (0.390)	−0.361 (0.341)	0.349 (0.283)
Project type: research (d)	−0.562*** (0.156)	−0.278** (0.118)	−0.392*** (0.090)
Total project budget (lnNOK)	0.179*** (0.053)	0.079* (0.044)	0.042 (0.050)
Energy (d)	−0.065 (0.130)	0.243* (0.134)	0.081 (0.069)
Engineering/ICT (d)	0.234 (0.173)	−0.214 (0.154)	0.179** (0.071)
Ocean (d)	−0.093 (0.152)	−0.146 (0.187)	0.065 (0.096)
Other (d)	−0.302 (0.183)	−0.083 (0.147)	−0.184* (0.103)
Years 2009–2011 (d)	0.368** (0.179)	0.419*** (0.140)	0.07 (0.117)
Years 2012–2014 (d)	0.453*** (0.113)	0.209* (0.110)	0.143* (0.078)
Constant	2.685*** (0.604)	3.101*** (0.554)	2.485*** (0.513)
Number of observations	484	484	484
R2	0.16	0.19	0.11
F(12, 44)/chi2	17.49	284.28	102.89
P-value	0.000	0.000	0.000
Rho		0.23***, chi2(1) = 25.231	

(d): dummy variable.

* p < 0.10.

** p < 0.05.

*** p < 0.01.

access agreements typically have a high diversity of partners involved, in contrast to proprietary partnerships, where concentration is high. In sum, the cluster analysis uncovers an important and sizable group of projects that do not fit an open-closed dichotomy but combine contractual provisions that are less industry oriented (regarding IP ownership) and more industry oriented (regarding access for commercial use or dissemination). In that sense, universities in these projects take on a rather entrepreneurial role: they own the foreground IP but hand out licenses for the commercial use of project results. In exchange, the dissemination of results is more restricted.

4.4. Interview results

In the following, we report the results of the interviews with seven project managers and participants of four projects in our sample dataset. We complement the interviews with information on the projects' outcomes gathered from the databases maintained by the RCN. We

Table 6
Descriptive statistics for the cluster solution.

	Cluster 1 (n = 189)	Cluster 2 (n = 205)	Cluster 3 (n = 90)	F-test
	“Proprietary partnerships”	“Controlled access agreements”	“Open science collaborations”	
Degree of industry ownership	4.058	1.985	1.967	***
Degree of industry use rights	3.487	3.259	2.278	***
Degree of publication restrictions	3.021	3.068	2.311	***
Degree of confidentiality restrictions	3.333	3.395	2.378	***
Share of RCN funding	0.525	0.636	0.641	***
Number of partners	7.143	8.683	9.044	***
Partner concentration	0.500	0.466	0.490	**
Project type: research (d)	0.132	0.151	0.311	***
Total project budget (lnNOK)	9.528	9.751	9.594	
Biotechnology (d)	0.323	0.327	0.378	
Energy (d)	0.291	0.337	0.178	**
Engineering/ICT (d)	0.122	0.093	0.133	
Ocean (d)	0.127	0.122	0.144	
Other (d)	0.138	0.122	0.167	
Years 2009–2011 (d)	0.138	0.098	0.022	**
Years 2012–2014 (d)	0.434	0.444	0.233	***
Years 2015–2017 (d)	0.429	0.459	0.744	***

(d) dummy variable.

** p < 0.05.

*** p < 0.01.

anonymised all identifying information and assigned each project a code name: “Health”, “Sustainability”, “Automotive”, and “Marine”, indicating its fields of research. Table 8 provides an overview of the four projects.

4.4.1. “Health”

The project “Health” is a collaboration between SMEs and universities on health technology, started in 2018 with a total budget of 20 MNOK and an RCN funding share of 50 %. The project's collaboration agreement stipulates joint ownership and non-exclusive use rights, broadly dispersing access to the project outcomes among the partners. At the same time, there are no publication restrictions and few requirements regarding confidentiality. In that sense, the project can be characterised as fairly open and resembling an “open science collaboration”.

In our interviews, we learned that the parties had a close relationship from previous collaborations. An SME manager told us: “We trusted the other parties to the extent that we did not use our lawyer to review the contract”. The initial collaboration agreement was not amended during the project period, and the results were used in developing commercial products and as the basis for further research. The project coordinator, a company, brought a strong and broad patent portfolio into the project as background IP. Since any follow-up use of the project results would, therefore, require access to the background IP already published in patent documents, the company was not concerned about sharing the project results as agreed with the other partners nor about publications.

Table 7
Regression results for the clusters of contractual provisions.

	Model 7 – Probit “Proprietary partnerships” (n = 189)	Model 8 – Probit “Controlled access agreements” (n = 205)	Model 9 – Probit “Open science collaborations” (n = 90)
Share of RCN funding	−1.736*** (0.397)	1.519*** (0.390)	0.192 (0.446)
Number of partners	−0.046*** (0.015)	0.016 (0.018)	0.035* (0.020)
Partner concentration	0.925** (0.431)	−1.124*** (0.309)	0.241 (0.498)
Project type: research (d)	0.142 (0.213)	−0.598** (0.245)	0.606*** (0.224)
Total project budget (lnNOK)	−0.126* (0.076)	0.162*** (0.057)	−0.043 (0.088)
Energy (d)	0.029 (0.168)	0.129 (0.166)	−0.315** (0.157)
Engineering/ ICT (d)	−0.12 (0.210)	0.014 (0.182)	0.116 (0.299)
Ocean (d)	−0.193 (0.179)	0.043 (0.180)	0.214 (0.213)
Other (d)	−0.032 (0.279)	−0.032 (0.277)	0.08 (0.138)
Years 2009–2011 (d)	0.766*** (0.261)	−0.03 (0.267)	−1.330*** (0.367)
Years 2012–2014 (d)	0.183 (0.114)	0.248* (0.128)	−0.611*** (0.165)
Constant	1.725** (0.689)	−2.293*** (0.672)	−0.812 (1.046)
Number of observations	484	484	484
Pseudo R2	0.10	0.07	0.12
Wald chi2(11)	97.83	63.61	49.03
P-value	0.000	0.000	0.000

(d): dummy variable.

* p < 0.10.

** p < 0.05.

*** p < 0.01.

As specified in the contract, the partners kept some details from the research results confidential to establish trade secrets underlying the commercialisation of the results. Moreover, one of the partners eventually decided to file a new patent application without giving a license to the others. However, the initial contractual provisions were not amended or re-negotiated, partly because the other partners had no commercial interest in the patented invention. The experienced project coordinator reflected on the industry partners in the consortium that were all SMEs: “We have many more conflicts if there are both large and small entities in a project.”

4.4.2. “Sustainability”

The project “Sustainability” was a broad collaboration between research institutes, universities, industry, and public bodies. It was started in 2016 with a total budget of 15 MNOK and an RCN funding share of 100 %. In the project's collaboration agreement, the partners agreed on joint ownership and non-exclusive rights as well as on no publication restrictions and few requirements regarding confidentiality, similar in openness to project “Health”. The project results were meant to be further used in sustainability-related projects, and there were no patent applications. Again, this project rather closely resembles an

Table 8
Overview of the interviews.

Project code name	Project and participant characteristics	Contractual provisions at project start	Project outcomes
“Health”	Share of RCN funding: 50 % Number of partners: 5 Share of companies: 3/5 Project type: commercial Total project budget: 20 MNOK Year of project start: 2018 Scientific field: biotechnology	Ownership to foreground IP: 3 Use rights to foreground IP: 3 Publication: 1 Confidentiality: 2	Patents, trade secrets, publications. Project outcomes mainly according to the initial collaboration contract.
“Sustainability”	Share of RCN funding: 100 % Number of partners: 10 Share of companies: 7/10 Project type: research Total project budget: 15 MNOK Year of project start: 2016 Scientific field: other	Ownership to foreground IP: 2 Use rights to foreground IP: 2 Publication: 1 Confidentiality: 2	Publications and results for further research. The project did not amend the contract but handled an unforeseen type of IP in accordance with the overall Open Science approach of the project.
“Automotive”	Share of RCN funding: 33 % Number of partners: 15 Share of companies: 9/15 Project type: commercial Total project budget: 30 MNOK Year of project start: 2016 Scientific field: engineering/ICT	Ownership to foreground IP: 2 Use rights to foreground IP: 4 Publication: 5 Confidentiality: 5	Publications, doctoral theses. While the handling of the IP and disclosure differed from the initial collaboration contract, no changes were made to the contract.
“Marine”	Share of RCN funding: 40 % Number of partners: 10 Share of companies: 6/10 Project type: commercial Total project budget: 25 MNOK Year of project start: 2016 Scientific field: biotechnology	Ownership to foreground IP: 5 Use rights to foreground IP: 5 Confidentiality: 5	Publications and results for further research. The provisions were re-negotiated when the project started, adding a publishing agreement.

“open science collaboration”. The project manager confirmed: “The project was fully funded by the Research Council, and since both universities and public entities were part of the consortium, the norm was full openness”.

The informants indicated that the initial contract remained unchanged. However, the research design was changed during the project, emphasising data collection from various Internet and other sources. Questions regarding the ownership of those data and privacy issues were

addressed in a process with the Norwegian National Data Protection Agency to make the data publicly usable for machine learning purposes. Although not foreseen in the collaboration contract, the high degree of trust among the participants and the involvement of public bodies facilitated the change in the project's direction without an amendment of the contract. Handling the data as a new IP type was per the project's overall open-science approach.

4.4.3. "Automotive"

The project "Automotive" was a large project involving 15 partners and a total budget of 30 MNOK. It was started in 2016 and coordinated by a subsidiary of a sizeable European industry group. Eight other participants were specialised Norwegian companies, joined by Norwegian research institutes and universities. The project received public funding for one-third of the total budget. While the collaboration contract stipulated joint ownership of the project results, the project coordinator had exclusive use rights, which were in a specific field. Moreover, results were not to be published and had to be kept confidential. In that sense, the project can be characterised as a "controlled access agreement".

Contrary to these provisions, however, it turned out that the other industry partners in the consortium had also made use of the results in their production processes. In addition, the project led to many scientific publications and engaged doctoral students. There were no patents applied for. In that sense, the actual outcomes of the project deviated to some extent from the ex-ante provisions, likely because they were process and not market related. The strict provisions regarding publication and confidentiality could have been used to enable the collaboration in the first place. Since the partners had not collaborated before, they had to build trust during the project, which later allowed them to deviate from the original provisions without changing the collaboration contract.

4.4.4. "Marine"

The project "Marine" was started in 2016 and involved ten participants, of which six were companies—direct competitors—besides universities and research institutes, with one of the research institutes being the project coordinator. The project had a total budget of 25 MNOK of which 40 % were publicly funded. The project was in a politically and ethically sensitive field. The collaboration contract stipulated that the industry partners would have ownership and exclusive use rights to all IP. Moreover, publications were banned, and all information had to be kept confidential. Hence, the project can be characterised as a "proprietary partnership".

It turned out that the project led to many scientific publications as well as follow-up projects based on the results. Our conversations revealed that the disclosure restrictions were initially adopted to facilitate an internal discussion among the partners to manage the ethical and political issues the intended results would touch upon. In that sense, they were provisional to get all industry partners on board, finalise the contract, and receive the funding agreed with the RCN. The project coordinator told us: "In the first project meeting, I presented a [separate] collaboration agreement that clearly stated ambitions for publications and a plan to include authors from all the partners. This was immediately accepted." This collaboration agreement was not managed as a contract amendment but was, in fact, a re-negotiation. It was facilitated by the trust that the partners had in each other from several prior collaborations. Hence, "Marine" is a project that most clearly deviates from the initial agreement. The project was based on "provisional confidentiality" until further details of the project could be agreed upon, after the support from the RCN was secured.

In sum, the ex-post interviews underscore that the initial collaboration contracts are necessarily incomplete and unable to account for any unforeseen circumstances during the projects' tenure. They may in some cases also be subject to re-negotiation, particularly regarding disclosure provisions, once uncertainty has resolved. However, our conversations also indicate that the projects' outcomes as well as the handling of the IP

and disclosure largely correspond with the initial collaboration agreements.

5. Discussion

Which contractual provisions governing IP and disclosure do university-industry collaborative research projects adopt, given the degree of public funding that the project receives and the competing orientations of the partners involved? Based on unique data from the collaboration contracts of 484 projects funded by the Research Council of Norway, our study highlights significant heterogeneity in how IP and public disclosure are governed, challenging a simple distinction between "open" and "closed" practices. Indeed, we find that controlled access agreements with ownership of the foreground IP allocated to universities and exploitation rights allocated to industry, combined with restrictions regarding publication and confidentiality of the project results, reflect the most common arrangement. At the same time, however, a substantial subset of projects adopts provisions that lead to proprietary partnerships, underscoring the tension between public goals of openness and industry's profit-driven objectives. But we also identify a sizable number of open science collaborations. This heterogeneity suggests that the specific context of each collaboration—such as the degree of public funding, the nature of the research, and the relative bargaining power of the partners involved—plays an important role in shaping the contractual terms. Moreover, the identified heterogeneity aligns with findings in prior literature on the competing orientations of academic and industry partners (He et al., 2021; Sauermann and Stephan, 2013), but also with recent observations of universities blending traditional norms of open science with commercialisation priorities (Rai and Cook-Deegan, 2017).

As expected, our findings confirm that public sponsorship of collaborative projects is associated with less concentrated ownership and exploitation rights, but it appears to have little association with disclosure practices. This qualifies prior research that public sponsors aim to promote open science principles by fostering broad dissemination of knowledge (Dasgupta and David, 1994; Mukherjee and Stern, 2009). Our results suggest that, for collaborative projects that may involve many different partners, the association between public funding and disclosure provisions becomes more elusive. For these provisions, our results show that the composition of the consortium matters more: a higher number of partners and, among those, more diversity in partner types are associated with less restrictive disclosure. Industry partners typically seek to protect proprietary interests, imposing restrictions such as publication bans or delays to keep results confidential to allow for a patent application or establish trade secrets (Czarnitzki et al., 2015b). Then again, a higher number of involved participants and less concentration of partner types seem to weaken the bargaining power for disclosure restrictions. Moreover, the partners may realise that disclosure restrictions are more difficult to enforce when the consortium is large.

Further, we find that larger projects in terms of total budget show a positive association with industry-oriented use rights and confidentiality restrictions. There may be several reasons explaining this observation. Larger projects typically involve greater risk, complexity and strategic importance, which may lead firms to prioritise the ability for commercial exploitation. But these factors may also lead universities to promote a greater degree of confidentiality. Indeed, prior research indicates that larger budgets lead university scientists to deny others access to their results and to be denied access to other scientists' work (Louis et al., 2001). Secrecy also depended on the scientists' entrepreneurial behaviour, that is, their engagement in commercialization, which may be higher in larger projects. These considerations would then be reflected in the contract negotiations.

The interviews with a selected group of project participants further deepen the insights from our analysis of the collaboration contracts. Here, we find that the contracts are typically not re-negotiated. At the

same time, the original contractual provisions did not hinder a re-negotiation. Still, the handling of IP and disclosure is adapted to the objectives of the participants as they develop a better understanding of the project, consistent with findings in prior research that point to evolving views on IP and disclosure in publicly funded innovation networks (Jarvenpaa and Wernick, 2011). However, these changes are not always formalised since the RCN's contract templates and the mandatory project reporting do not encourage a review of the contract provisions. Our findings highlight the importance of social processes among the participants that complement the legal provisions (Bok, 1982; Costas and Grey, 2014), as well as the use of various management styles (Coali et al., 2024). Re-negotiations are often informal, for example, in the case where the partners agreed to less restrictive disclosure provisions after the contract had been signed and in the case where a new type of IP emerged during the project.

Our results reveal a shift toward more open governance practices over time. Since 2009, the degree of industry orientation of projects in the RCN portfolio has continuously declined, indicating that the governance of project results increasingly aligns with the norms of open science. While this development is not due to a shift in policy at the RCN, it broadly reflects the intentions of the RCN that public funding should enable a broad dissemination of knowledge, as our conversations with the RCN indicate.⁸ At the same time, the results of the cluster analysis also show clear differences in the three profiles for project governance, with most projects adopting controlled access agreements.

In that sense, our study makes two primary contributions to the literature on university-industry collaborations. First, we focus on collaboration agreements as representations of the negotiated intentions of project participants regarding the governance of research outcomes. Building on earlier research that describes these agreements as intricate legal frameworks governing the IP and the disclosure of research results (Bstieler et al., 2014; Egelie et al., 2019; Gans et al., 2017), we document a large degree of heterogeneity in the way these contractual provisions are designed and, consequently, how and to what extent publicly funded research results enter the public domain. By introducing a novel approach for assessing collaborative research projects ex-ante, our findings contribute to the ongoing debate on open science in universities' engagement with industry that so far has exclusively considered the outcomes of those projects (e.g., Czarnitzki et al., 2015b; Perkmann et al., 2013) but not the legal starting point.

Furthermore, concerns are growing about knowledge monopolies driven by concentrated ownership and secrecy (Rai and Cook-Deegan, 2017). We demonstrate how collaboration agreements facilitate the early identification of emerging IP and disclosure practices, providing policy makers with an early and improved understanding of how projects seek to govern the IP and disclosure of research results. In that sense, our research also makes a methodological contribution by systematically coding and analysing contractual provisions. The approach is replicable for future research to explore similar dynamics in other contexts. The recent development of large language models and artificial intelligence may facilitate the practical analysis of contract portfolios in terms of their types of provisions (Dahlander et al., 2021; Egelie et al., 2019; Van Overwalle, 2013), for example, for public funding bodies, universities and corporations that seek to foster open science.

Second, we advance our understanding of how public sponsorship and the partners' competing orientations shape the contractual provisions of collaborative projects. The extant literature on research

sponsorship has often assumed a rather simplistic view of public sponsors promoting openness and private sponsors enforcing restrictions to allow for research results to get commercialised (David, 2004; Gans and Murray, 2012). Our results show that public sponsorship of collaborative projects and the variety of objectives and motivations among the participants interact non-trivially, shaping a nuanced set of contractual provisions.

Hence, our results hold several implications for policymakers in aligning public funding with broader policy goals. The ex-ante analysis of contractual provisions enables gaining detailed insights into the projects' governance of IP and disclosure, and thus developing strategies for managing research project portfolios (Howoldt and Borrás, 2023). Contract templates can be designed to support certain desired provisions to ensure that they align with policy goals as the projects evolve. For example, they can allow for re-negotiations to increase a project's openness once the partners have built trust and uncertainty about the likely project outcomes decreases. In that sense, the templates should reflect that collaboration contracts are necessarily incomplete and that the project outcomes are unknown ex-ante.

Moreover, the observation that higher shares of public funding are associated with how contractual provisions are devised suggests a potential leverage point for public sponsors to influence the governance of collaborative research. Policymakers could consider tying funding to specific contractual requirements that align with open science objectives, mandating non-exclusive licenses or requiring public disclosure within defined timeframes without stifling the commercial potential that motivates industry participation. Such measures could mitigate the risk of knowledge monopolies, as exemplified by cases like the CRISPR-Cas9 patent disputes (Egelie et al., 2016; Rai and Cook-Deegan, 2017). The Open Science Partnership (OSP) model, which aims to secure access for all, is a case in point (Gold, 2021). However, the extent to which such models are effective critically hinges on the ownership of background IP that large patenting entities might control and that may not be shared within the partnership (Norm et al., 2022), allowing a partner with relevant background IP to effectively limit the extent to which other parties can benefit from project results (Norm et al., 2024).

Finally, our results inform current debates at the EU level on Open Strategic Autonomy, highlighting how funding agencies and policymakers need to reconcile goals of knowledge diffusion and openness with concerns about technological sovereignty and control over research outputs (Edler et al., 2023). Our research suggests that the contractual provisions governing IP and disclosure can play a key role in balancing these conflicting objectives by creating trusted environments for collaboration.

6. Conclusion

This research investigates how public funding and competing orientations shape the contractual provisions regarding IP and disclosure in university-industry collaborations. Based on the contracts of 484 collaborative projects, we document considerable heterogeneity in how these provisions are designed as well as how they depend on the degree of public funding and the composition of the consortium. We demonstrate how contractual negotiations reflect the strategic interests of universities and industry partners, and how the outcome governs IP and disclosure of project results. We identify three distinct patterns of contractual provisions—"Proprietary partnership", "Controlled access agreements", and "Open science collaborations"—in response to our research question. Moreover, we find that the degree of public funding is negatively associated with the degree of industry ownership and use rights, but not with publication and confidentiality restrictions. Proprietary partnerships (controlled access agreements) exhibit a share of public funding below (above) the average.

While providing unique insights from data that are otherwise inaccessible for research purposes, our research is not without limitations, which in turn offer opportunities for future research. In particular, while

⁸ We have studied the RCN webpages and annual reports for the relevant period. We can only find a policy-shift regarding "Open Access" publishing and "Plan S". (There are later policy-shifts from RCN). We speculate that the observed development could be associated with the increasing professionalisation of the technology transfer function at the Norwegian universities, following the legislation change in 2003 where the professors' privilege was repealed.

we capture the initial agreements, the interviews suggest that modifications may occur as projects unfold, influenced by shifting priorities, challenges, and opportunities. These may affect the eventual governance of the research outcomes. The modifications could also differ for different types of IP: copyright, databases, patents and trade secrets could, for example, be the control mechanisms for AI-related research. As patent applications are published but other rights could remain secret, the governance could develop over time. Therefore, one promising direction for future research is to investigate how the contractual provisions identified here evolve over the lifecycle of a project and to systematically link the decision-making and actual outputs to the contractual provisions that the parties initially agreed on. Relatedly, further research is needed to examine the long-term societal and economic impacts of the contractual provisions studied here. Understanding how these agreements influence the dissemination and application of research outcomes can provide deeper insights into the effectiveness of public funding strategies.

Moreover, our research is limited in that it is based on a portfolio of projects funded by the Research Council of Norway and governed by Norwegian law. Norway uses a civil-law system (such as in the EU and China), where the background law regulates more issues than agreements in a common-law system (such as in the UK and the US). Consequently, the contracts under study may have been less detailed than in a different context, limiting the generalizability of our results. Hence, future research could seek access to collaboration agreements in various countries for comparative analysis.

By uncovering the fine print of collaboration, our study highlights the diverse ways contractual provisions govern IP and disclosure in publicly funded research. We hope it inspires further exploration of how agreements evolve and impact societal outcomes, ultimately shaping more equitable and effective policies for public research funding.

Geolocation information

Our study uses data from Norway.

CRediT authorship contribution statement

Haakon Thue Lie: Writing – review & editing, Writing – original

Appendix A

Table A1
Probit regression results.

	Model 1	Model 2	Model 3	Model 4
	Industry ownership	Industry use rights	Publication restrictions	Confidentiality restrictions
Share of RCN funding	−3.103*** (0.323)	−1.854** (0.758)	0.202 (0.227)	−0.460 (0.379)
Number of partners	−0.111*** (0.035)	−0.131*** (0.041)	−0.039** (0.016)	−0.039*** (0.015)
Partner concentration	0.515 (0.428)	−0.783 (0.891)	0.985* (0.559)	1.384*** (0.373)
Project type: research (d)	−0.118 (0.413)	−0.214 (0.314)	−0.223 (0.161)	−0.026 (0.202)
Total project budget (lnNOK)	−0.023 (0.077)	0.1 (0.096)	0.072 (0.092)	0.215*** (0.078)
Energy (d)	0.423* (0.247)	0.032 (0.354)	0.331*** (0.110)	−0.239** (0.115)
Engineering/ICT (d)	−0.573** (0.239)	−0.831*** (0.297)	0.251 (0.201)	−0.092 (0.182)
Ocean (d)	−0.148 (0.272)	−0.464 (0.433)	0.245* (0.139)	−0.357** (0.140)
Other (d)	0.095 (0.259)	−0.068 (0.305)	−0.125 (0.165)	−0.408** (0.176)

(continued on next page)

draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Knut Jørgen Egelie:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Christoph Grimpe:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Roger Sørheim:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Haakon Thue Lie reports financial support and travel were provided by Dehns. Knut Jørgen Egelie reports financial support and travel were provided by NTNU Technology Transfer AS. Haakon Thue Lie reports a relationship with Dehns that includes employment. Knut Jørgen Egelie reports a relationship with NTNU Technology Transfer AS that includes employment. The other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table A1 (continued)

	Model 1	Model 2	Model 3	Model 4
	Industry ownership	Industry use rights	Publication restrictions	Confidentiality restrictions
Years 2009–2011 (d)	–0.521 (0.362)	0.263 (0.287)	–0.222 (0.240)	0.931*** (0.243)
Years 2012–2014 (d)	–0.279 (0.173)	–0.07 (0.178)	–0.035 (0.127)	0.716*** (0.124)
Constant	1.401** (0.641)	–0.071 (1.335)	–1.800** (0.902)	–2.820*** (0.846)
Number of observations	484	484	484	484
Pseudo R2	0.24	0.17	0.04	0.11
LR/Wald chi2	480.95	708.134	40.53	113.453
P-value	0.000	0.000	0.000	0.000
Log likelihood	–150.096	–97.733	–253.074	–275.203

(d): dummy variable.

* $p < 0.10$.** $p < 0.05$.*** $p < 0.01$.**Table A2**

Semi-structured interview guide.

The questions were sent ahead of the interview and covered:

- Background for the project
 - Do you know if any background IP that was needed for the utilisation of the results from project was patented or kept as a trade secret? Please explain.
- Results from the projects (in addition to those listed in Prosjektbanken [RCN's overview of project grants and results])
 - Are you aware of any patents resulting from the project
 - Are you aware of any confidential results, like trade secrets?
 - Were any companies formed based on the research results?
 - Were any new R&D collaborations founded based on the research results?
- Negotiation of the contractual provisions
 - How were the contractual provisions negotiated, were they amended or re-negotiated, were there were any conflicts?
 - How did the parties discuss and agree on the openness and confidentiality before, during and after the project?

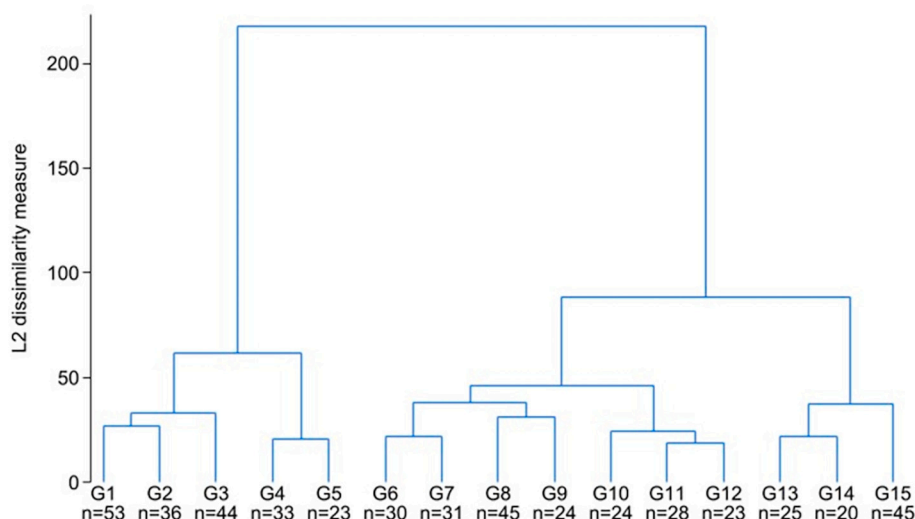


Fig. A1. Dendrogram of the cluster solution (Ward's linkage, Euclidean distance).

Data availability

The regression results were obtained using STATA SE version 19 statistical software. The do-file can be obtained from the authors upon request. Our copies of the project contracts have been shredded in accordance with the agreement with the Norwegian government.

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